

## ★ X-Wing

An X-Wing has exactly two possible positions for one digit in each of two rows, lined up in the same two columns. The two copies occupy opposite corners—either diagonal is possible. Both choices put one copy in each column, so remove the digit from the other cells in those columns.

The pattern also works sideways: two columns confine the digit to two rows. Then remove it from the rest of those rows. The starting rows or columns are the *base lines*; the crossing ones are the *cover lines*.

### Example: X-Wing

Track candidate 8 in Rows 5 and 8. In each row it can appear only in Columns 3 and 7. The four blue corners form an X-Wing.

All base-row positions for this digit are marked. The notes are complete; follow this digit across them. We show one useful elimination; there may be others.

**Check the other positions.** For 8 in Row 5: R5C4, R5C5, and R5C6 see the 8 at R6C5. For 8 in Row 8: R8C2 sees the 8 at R2C2; R8C5 sees the 8 at R6C5; R8C6 sees the 8 at R9C4.

|    | C1 | C2 | C3               | C4 | C5 | C6 | C7          | C8 | C9 |
|----|----|----|------------------|----|----|----|-------------|----|----|
| R1 | 5  |    | 9                |    |    | 2  | 4           |    |    |
| R2 | 7  | 8  |                  |    |    | 9  |             |    | 6  |
| R3 |    |    | 6                |    |    |    |             |    |    |
| R4 |    |    | 5<br>8           |    | 3  | 1  |             |    |    |
| R5 | 4  | 7  | 3<br>8           |    |    |    | 2<br>6<br>8 | 5  | 1  |
| R6 |    |    |                  | 4  | 8  |    |             |    |    |
| R7 |    |    |                  |    |    |    | 9           |    |    |
| R8 | 6  |    | 1<br>4<br>5<br>8 | 9  |    |    | 5<br>8      | 2  | 7  |
| R9 |    |    | 2                | 8  |    |    | 1           |    | 3  |

The two rows need two copies of the digit. They must occupy different cover columns, so each cover column receives one.

### Next Step: Eliminate and Place

The X-Wing claims candidate 8 in both cover columns. Remove 8 from R4C3 outside the two base rows. Its candidates change from {5, 8} to {5}.

|    | C1 | C2 | C3                         | C4 | C5 | C6 | C7                         | C8 | C9 |
|----|----|----|----------------------------|----|----|----|----------------------------|----|----|
| R1 | 5  |    | 9                          |    |    | 2  | 4                          |    |    |
| R2 | 7  | 8  |                            |    |    | 9  |                            |    | 6  |
| R3 |    |    | 6                          |    |    |    |                            |    |    |
| R4 |    |    | 5                          |    | 3  | 1  |                            |    |    |
| R5 | 4  | 7  | <div>3<br/>8</div>         |    |    |    | <div>2 3<br/>6<br/>8</div> | 5  | 1  |
| R6 |    |    |                            | 4  | 8  |    |                            |    |    |
| R7 |    |    |                            |    |    |    | 9                          |    |    |
| R8 | 6  |    | <div>1 3<br/>4<br/>8</div> | 9  |    |    | <div>5<br/>8</div>         | 2  | 7  |
| R9 |    |    | 2                          | 8  |    |    | 1                          |    | 3  |

Place 5 in R4C3. This single was not available before the X-Wing; now rescan its row, column, and box.